

GreenThreads (Sustainable Apparel) | Finance Function

AI.205: AI Integration in Business I | Homework #1: Functional Brief

Team: Kyara (Keke) Doble, Sahreyah Herring-Jones, Will Pierre, Sherman Cole

1. Context

For context, the company we are working at is GreenThreads, a Series B sustainable apparel retailer with 12 stores and two channels, retail and e-commerce, with approximately \$40M in revenue and not yet profitable. The function we serve at the company is the finance team.

2. The Finance Function at GreenThreads

GreenThreads' finance function is responsible for stewarding the company's money from plan to proof. The finance function owns the annual and quarterly budgets, which includes tracking day to day expenses against the budget. The finance department also plans and monitors the fixed and variable costs that drive the business. This includes store rent or purchasing a building outright, the buildout of new locations, and sourcing materials for the company, such as organic cotton and bamboo textiles. The function processes and records transactions and manages payables and receivables. Because GreenThreads is expansion focused, Finance also carries the forward looking work: modeling the cost of new store openings, reviewing whether revenue and margin targets are being hit, and flagging variances early enough for leadership to act on them.

The function serves the company as a whole, but its most direct customer is the CFO. The CFO relies on Finance to produce accurate, timely reporting for the executive team and the board. Since the Series B raise, the board expects financial discipline. Finance is the function that translates activity in stores, marketing, and the supply chain into numbers investors can trust. Internally, Finance supports every other department by setting budgets, approving spend, and answering the question every team eventually asks: can we afford this, and is it working?

Success in the finance department is measured by our ability to stay at or under budget. Coming in under budget saves the company money, but the deeper measure is whether the dollars we do invest generate more money for the business. If we spend less while the spend we approve drives revenue, Finance is doing its job. Success also means healthy cash management, with strong cash flow coming through, receivables collected on time, and adequate runway maintained for a company that is funded but not yet profitable. Finally, success means delivering proper quarterly reporting to our investors so

they know how the company is performing.

3. Workflow Map

The table below documents the function's typical daily, weekly, and quarterly tasks.

Frequency	Task	Purpose	Inputs	Outputs
Daily	Check emails and messages	Respond to requests and issues	Emails, team messages	Responses, updated tasks
Daily	Process transactions/data	Keep records current and accurate	Customer requests, reports	Updated records
Daily	Attend team meetings	Share updates and priorities	Team agenda	Action items
Weekly	Review performance metrics	Monitor progress toward goals	Reports, dashboards	Recommendations and adjustments
Weekly	Complete status reports	Communicate progress	Work completed during week	Weekly report
Weekly	Resolve outstanding issues	Prevent delays and bottlenecks	Open tickets/tasks	Closed issues
Quarterly	Analyze department performance	Evaluate effectiveness and results	Quarterly data and reports	Performance summary
Quarterly	Budget and resource planning	Prepare for upcoming needs	Financial and staffing data	Updated budget/resource plan
Quarterly	Process improvement review	Identify opportunities for efficiency	Employee feedback, performance data	Improvement plan

4. AI Opportunities: Identification and Classification

Grounding note: We pulled the MIT Sloan framework from the assigned article, which describes three levels of AI value. Level 1 boosts individual productivity (inbox, meetings, briefings, writing tone). Level 2 incorporates generative AI into well-defined tasks and roles, where developers get help writing code, analyzing data, and creating documentation, and AI agents provide sales and call center representatives with fast answers to common questions. Level 3 brings automation to production and operational

with fast answers to common questions. Leverage brings automation to production and operational processes, where AI can assist teams in creating entire campaigns and enterprise software can automate processes like managing supply chains or identifying workforce skills gaps.

Opportunity Inventory

#	Task	Level	Why it fits
1	AI-drafted budget-versus-actual variance flagging, giving analysts a first-pass list of lines that moved instead of a blank spreadsheet	2	Well-defined analytical task, human still signs off; matches the framework's data-analysis-in-a-role pattern
2	AI-drafted variance narratives for the flash P&L and board package, with the analyst editing rather than writing from scratch	2	Structured commentary against source-system numbers with clear verification criteria
3	Inbox and meeting-notes triage for the finance team (summarizing vendor and audit calls, flagging action items)	1	Textbook individual-productivity use case per the framework
4	AI-assisted first drafts of recurring finance communications, such as spend-policy reminders and budget-cycle instructions to department heads	1	Same business-writing use case the framework cites for tone and clarity
5	Internal knowledge agent answering policy questions from other departments, such as spend approval thresholds and expense coding	2	Directly mirrors AI agents giving representatives fast answers to common questions

6	Automated invoice matching and proposed payment batches, tied to purchase orders and receiving data, running with minimal human review	3	Full operational process automation; highest effort, highest risk
7	Automated end-of-quarter ESG and sustainability reporting pack (pulling supplier, shipping, and materials data into a drafted report)	3	Multi-source production process, not a single task

Ranked by Impact and Feasibility

Rank	Opportunity	Feasibility	Impact	Rationale
1	#2 Flash P&L variance narratives	High	High	Narrow, structured, checkable against source data, and already gated by our local rule in section 5

2	#1 Budget-versus-actual variance flagging	Medium	High	Big close-cycle time savings, needs clean data feeds
3	#3 Inbox/meeting triage	High	Medium	Fast to stand up, steady but modest time savings
4	#5 Internal policy knowledge agent	Medium	Medium	Needs a maintained policy knowledge base first
5	#4 Finance communications drafting	High	Medium	Easy win, tone and accuracy need a QA loop
6	#7 ESG reporting automation	Low	High	High payoff but multi-source integration is heavy
7	#6 Automated invoice matching and payment batches	Low	High	Highest long-run value, but wrong first move; segregation of duties requires Level 1/2 trust built first

Top Opportunities, Quantified

#1: Flash P&L variance narratives (Level 2). Assume the monthly close covers roughly 30 budget lines and an analyst spends about 10 minutes per line drafting commentary on what moved and why, or about 5 hours per close, plus additional drafting for the quarterly board package. An AI-generated first-pass narrative that the analyst verifies against source-system numbers and edits, rather than writes from scratch, could plausibly cut drafting to about 2 hours per close: roughly 3 hours saved per month and closer to 6 in board-package months, with more consistent commentary across periods. Alignment in one sentence: it speeds up the accurate, timely reporting to leadership and investors that our function is measured on, without loosening the sign-off gate our local rule requires.

#2: Budget-versus-actual variance flagging (Level 2). If analysts currently spend about 4 hours per week reviewing spend transactions against budget lines to catch overruns, an AI first-pass that flags the lines that moved most, with the analyst confirming each flag, could plausibly cut review to about 90 minutes: roughly 10 hours saved per month, and more importantly, a consistent flagging pass reduces the odds of an overrun going unnoticed until quarter end, which is the costlier failure mode. Alignment in one sentence: it directly supports our core success measure of staying at or under budget by surfacing variances early enough for leadership to act on them.

A caution worth carrying into any pitch: per the article, hallucinations cannot be eliminated, so structured tasks and data formats that are easier to validate are the safer place to start, which is exactly why #2 and #1 (both structured, checkable outputs) rank ahead of the harder-to-validate #6 and #7.

5. Guardrails and Local Rules

Risk	Category	Rule / Mitigation
Sensitive financial and banking data exposed to a public AI tool	Executive guardrail	Company-wide policy: no bank account numbers, customer payment data, or unredacted contracts may be pasted into a non-approved AI tool. Only the enterprise-licensed, access-controlled LLM instance may touch raw financial data.
AI-drafted numbers or narrative reaching leadership without verification (hallucination risk)	Local rule (Finance function)	Any AI-generated commentary, forecast, or figure used in the flash P&L, board package, or ESG report must be traced back to source-system data and signed off by the preparing analyst before distribution. No AI output goes out ungated.
Loss of segregation of duties if AI is allowed to both draft and approve transactions	Executive guardrail	AI tools may recommend or draft (for example, propose a payment batch or flag a collections priority), but a human with existing payment authority must approve. AI is never granted standalone approval or release rights in the payment system.

6. Tools and Verification

To pressure-test this brief, we used Claude (a fast, everyday-tier model) to sanity-check whether our classification of each AI opportunity matched the MIT Sloan Level 1/2/3 definitions as described in the assigned reading, and separately used a more powerful model tier to help draft the variance-narrative example so we could see what realistic AI-generated financial commentary looks like. We verified the level classifications by re-reading the MIT Sloan article's own examples side by side with ours, and we checked the drafted variance narrative against a mock set of budget-versus-actual numbers to confirm it did not invent figures that were not in the source data, consistent with only using AI on material we understand well enough to catch its mistakes.

Sources

Mollick, E. "Using AI Right Now." Assigned Week 1 reading.

MIT Sloan Management Review. Assigned Week 1-2 reading on scaling AI (Level 1/2/3 framework).

AI.205 GreenThreads Case Brief, Capstone 1, Summer 2026.

